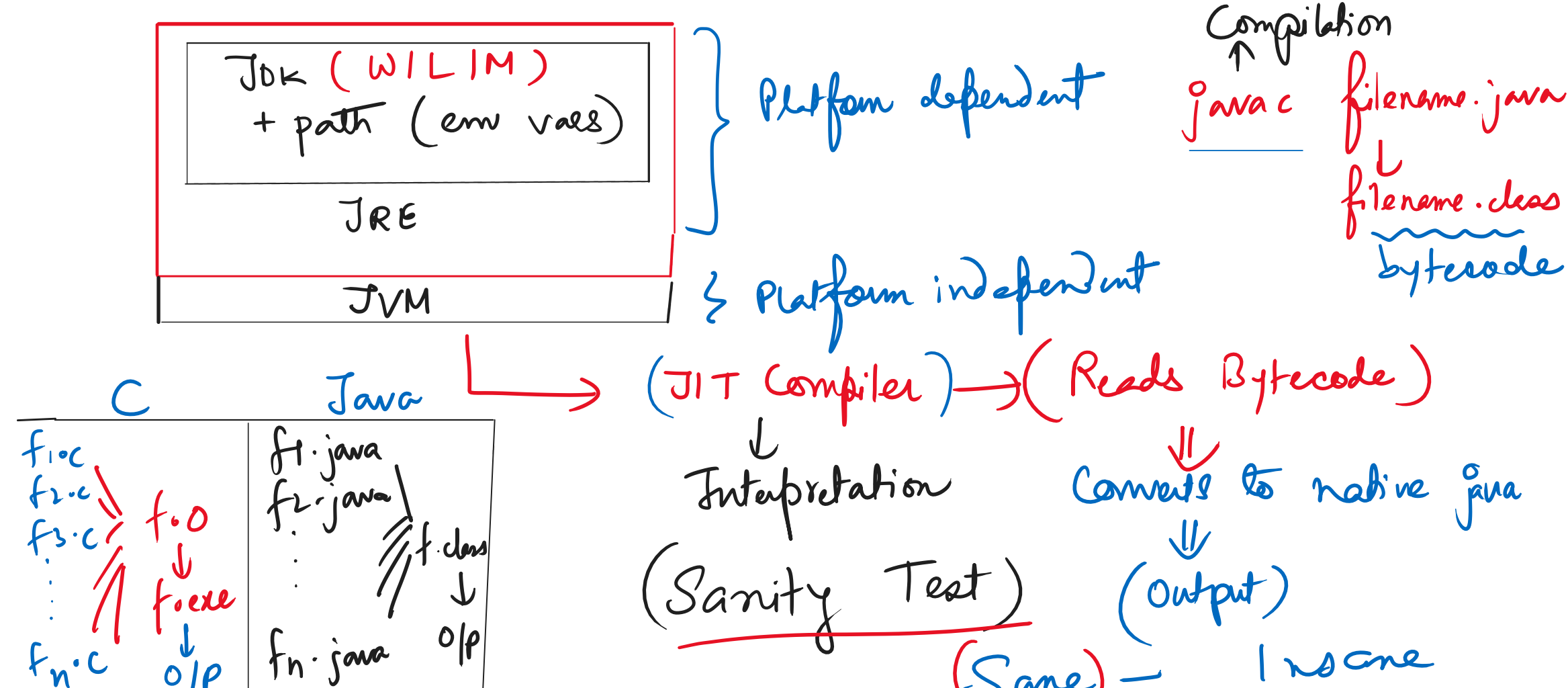




Data Types in Java: $\Rightarrow$

(Primitive)
(Simple Values)

Not a pure OOL language

- byte
- short
- int
- long
- float
- double
- char
- boolean

(Complex Objects)
Non-Primitive

Reference Data Types

All Classes
String
Date
Scanner Class

Operators: ① Arithmetic Operators: $+, -, /, *, \% [//, **]$

② Assignment: $=, +=, -=, /=$

③ Logical: $\&\&, \&\&, !$ (and or not)

④ Relational / Comparison: $>, <, >=, <=, ==, !=$

⑤ Ternary operator: (condition) ? tv : fv;

Java: return-type var = (condition) ? tv : fv; (C/C++)

* Unary Operators: $++$ / $--$

* Bitwise: ⑥ Individual Bits

AND $\&$
OR $|$
XOR $\wedge$
RS $\gg$
LS $\ll$
NOT $\sim$

0111 7
1000 8
0000 = 0
int & int = int

(2 > 3) && (3 > 4)
false && false

0101 5
1001 9
1101 = 13
5 | 9 = 13

S = 0
D = 1

0110 6
xor 0011 3
0101 = 5

Bit Masking ***
or
Bit Manipulation
value, unit/step

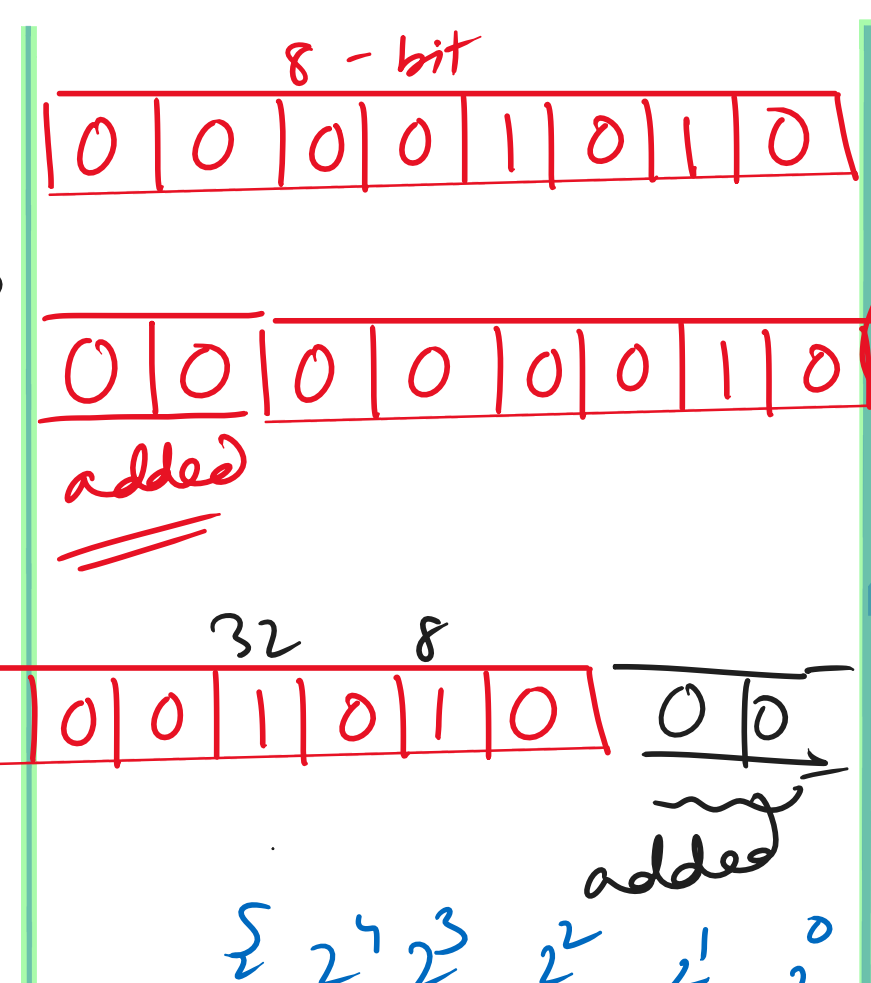
a = 10

a << 2

10 << 2

= 40

discarded



a >> 2

10 >> 2

= 2

discarded

Bitwise

NOT

Reverse

0 1
1 0

T F
F T

(~S = -6) (PV)

① (-6 = 10)

abs(-6) = 6 $\rightarrow$ 0110

(Negation of 2) $\left\{ \begin{array}{l} 1's \text{ com} \\ +1 \end{array} \right\}$

2's Com

(PV) General Rule: $\sim n = -n - 1$

500 = -501

-500 = -(-500) - 1 = -499

S $\rightarrow$ 0101

~S $\rightarrow$ 1010 = 10

TV

(MCQ)

~S

5

6

10 x

-6 x

(Power of Two) ⑦

7 x

0111

0110

not 0

11

4 x

0100

0011

yes

8 x 7

1000

0111

8 x yes

if (n & (n-1) == 0)

{

pow2;

else {

not pow2;

}

9 x

1001

1000

No

0110

0101

6 x

No

* Hamming Weight: $\Rightarrow$

LC(191) No of 1 bits

int hw (int n) { while

{

11 $\rightarrow$ 1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

0001

1

T

1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

001

1

T

1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

01

1

F

1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

1

T

1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

0

1

T

1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

1

T

1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

1

T

1011 $\gg 1 = 101 \gg 10 \gg 1 \gg 1$ Count = 4

1

T

11 $\rightarrow$ 1011

$\rightarrow$ ③

5 $\rightarrow$ 0101

$\rightarrow$ ②

0 $\rightarrow$ 0000

$\rightarrow$ ①

Count = 4

0 $\rightarrow$ stop

1

2

3

TCQ

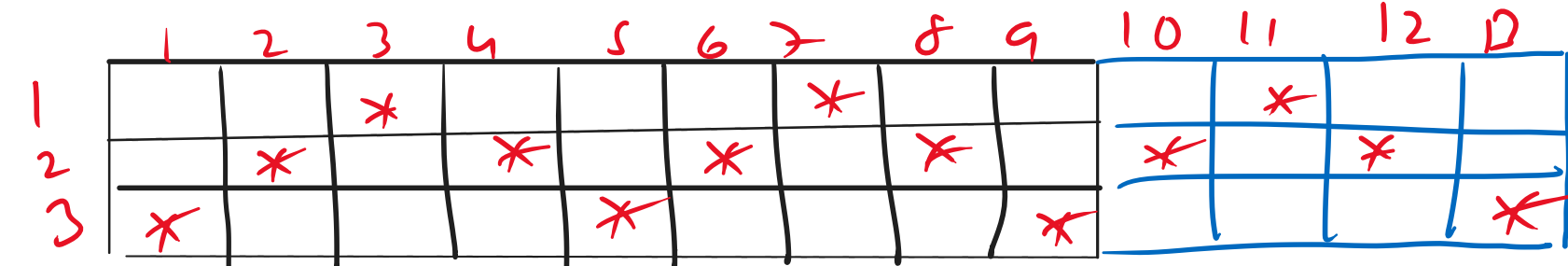
n > 0 n != 0

r = 3

c = 9, 13, 17, 21

Zig Zag Pattern

Time is less



R1 3, 7, 11 = C/4 = ③ ①

R2 Even No = C/2 = ②

R3 1, 5, 9, 13 = C/4 = ①

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

1 2 3 4 5 6 7 8 9

Pyramid n=5

spaces stars

n-i 2x i-1

for (1 to n)

{

for (space)

n-i

for (stars)

2x i-1

}

0(n²)